

Jake Gendreau

Advanced PLC Programming

Dan Blanchette

April 27, 2026

Lab 4 Report

My lab 4 implementation has four distinct parts. Those parts are the Web Page, the API, the Pylogix communication, and the systemd script.

- Web Page
 - Hosted with Django, utilizes basic HTML + CSS to display information. Variables are filled at render time with Django templating.
- API
 - An endpoint that is accessed through the systemd script. It calls the Pylogix script to scan and update tags, then updates the database for use in the frontend.
- Pylogix Script
 - The Pylogix Script has two functions. One reads and returns the two relevant tag values, and the other increments each of the tag values.
- Systemd Script
 - The systemd script runs every 10 seconds and uses cURL to access the API, which causes the pylogix script to update the variables.

Since I have been using Linux as a daily driver for two years, I did not have much difficulty with this lab. The hardest part was reading the rules. I implemented it initially by having the values get read from the PLC when the page has a GET, but I had to redo that so that it would read from the database.